Solutions and teaching notes for chapter 3

Teacher use. The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $(3, -4)$. *Action:* Subtract end minus start on two drawings.
2. $\sqrt{3^2 + (-4)^2} = 5$. *Action:* Review Pythagoras and distinguish norm from coordinates.
3. $\overrightarrow{AC}$. *Action:* Draw the composite displacement.
4. $a^2 + 2ab + b^2$. *Action:* Identify the cross term.
5. $\frac{3}{5}$. *Action:* Name the sides relative to the angle.
6. $(1, 2)$. *Action:* Average the two coordinates separately.

Investigation activity

Take points on the circle with diameter AB; proof via the midpoint I: product $MI^2 - 9$. The endpoints are included because one of the vectors is zero.

Immediate applications

1. 0 then -10. The cosine is 0 and -1 respectively.
2. $3t - 12 = 0$, so $t = 4$.
3. The square is $4 - 2 + 9 = 11$; the norm is $\sqrt{11}$.
4. $32 = 16 + 16 - 32\cos A$, so $\cos A = 0$ and $A = 90°$.
5. $MI^2 - 9 = 7$: circle with centre $(3, 0)$ and radius 4.

Exercises: method and results

Exercise 3.1 – Direct calculations. Product $4 - 6 = -2$; norms $\sqrt{5}$ and 5; sum $(5, 1)$ with square 26. Check: $5 + 2(-2) + 25 = 26$.

Exercise 3.2 – A right triangle? $\overrightarrow{AB} = (3, 1)$, $\overrightarrow{AC} = (-1, 3)$; product 0. $AB = AC = \sqrt{10}$, area $\frac{AB \times AC}{2} = 5$ area units.

Exercise 3.3 – Length without a frame. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{AD}$ and $\overrightarrow{BD} = \overrightarrow{AD} - \overrightarrow{AB}$. Product $\frac{4 \times 3}{2} = 6$; $AC^2 = 16 + 9 + 12 = 37$ and $BD^2 = 16 + 9 - 12 = 13$: the diagonal $[AC]$ is the longest, which corresponds to an acute angle $\widehat{BAD}$ (positive dot product).

Exercise 3.4 – Recovering an angle. $\cos\theta = -\frac{6}{12} = -\frac{1}{2}$; in $[0, 180°]$, the angle is 120°. The domain ensures uniqueness.

Exercise 3.5 – A variable locus. $MI^2 = k + \dfrac{AB^2}{4}$. If the right-hand side is negative: empty set; zero: the single point I; positive: circle whose radius is its square root. The discussion bears on a squared distance.

Exercise 3.6 – Work done by a force. $W = \dfrac{50 \times 4 \times 1}{2} = 100$ J. At 120°, $W = -100$ J. The sign distinguishes help to the displacement from opposition; it does not describe a negative distance.

Exercise 3.7 – Contrapositive. The contrapositive is "if $\vec{u}$ and $\vec{v}$ are orthogonal, then $\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2$"; it follows from the expansion, since $\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u}\cdot\vec{v} + \|\vec{v}\|^2$ and $\vec{u}\cdot\vec{v} = 0$. It is equivalent to the original statement. We can also write $2\vec{u}\cdot\vec{v} \neq 0$.

Exercise 3.8 – Delayed review. Al-Kashi: $\dfrac{9+16-25}{2} = 0$. Converse of Pythagoras: triangle right-angled at A, so zero product. Name the hypotheses of each method.

CQFD, exercise 9 p. 250, questions a and b. a: $1 \times 3 + 2 \times (-1) = 1$. b: $(-1) \times 12 + 3 \times 7 = 9$. The scalar product is a number, not a vector.

CQFD, exercise 11 p. 250, questions 1, 2 and 3. $\overrightarrow{AB} = (4,6)$, $\overrightarrow{AC} = (-3,4)$, $I(-1,-2)$ and $\overrightarrow{CI} = (5,-1)$. Hence $\overrightarrow{AB}\cdot\overrightarrow{AC} = 12$, $\overrightarrow{AB}\cdot\overrightarrow{CI} = 14$. Chasles gives $\overrightarrow{AB}\cdot\overrightarrow{AI} = 12 + 14 = 26$. Two checks: $\overrightarrow{AI} = (2,3)$ gives $8 + 18 = 26$; or $\overrightarrow{AI} = \dfrac{1}{2}\overrightarrow{AB}$ gives $\dfrac{1}{2}(16+36) = 26$.

MCQ: answers and reasons

MCQ 1.

1. A. $6 - 4 = 2$.
2. C. Square root of 25.
3. D. Scalar criterion.
4. B. $\dfrac{2 \times 3}{2} = 3$.
5. C. Expand the scalar square.

MCQ 2.

1. D. Negative cosine.
2. A. Coordinates by projection.
3. C. $MI^2 = \dfrac{AB^2}{4}$.
4. B. Bilinearity: $2(-1)3$.
5. D. $BC^2 = 8$.

True or false

1. False: $u \cdot (-u) = -\|u\|^2 < 0$ for $u \neq 0$.
2. True: its product with every vector is 0.
3. False: $(1,0)$ and $(0,1)$ are nonzero and orthogonal.

Differentiated teaching. For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.